IMDB Midterm Project

MGCS-661 Section 075
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McGill

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INTRODUCTION

Magic is fascinating to humans. From ancient tales of witchcraft and wizardry to the modern 'movie magic' humans persistently seek out the impossible. One of the most prevalent feats of magic is looking into the future. For this project, the team will cast a spell to predict the future.

Specifically, the team is looking to predict the IMDB ratings of 12 upcoming movies. Based on an IMDB dataset of 2,000 previously released movies, the team will generate a prediction for each of the 12 upcoming movies selected. The goal is to create a prediction model using R studio that will accurately predict the ratings of these movies. The accuracy of a prediction is made by comparing the prediction with the actual value when the movie is released. Therefore, if our crystal ball predicts a rating of 4.7 and the actual rating is 4.7 then the crystal ball is of high quality.

Magicians never reveal their secrets… except to other magicians. The challenge with the prediction model is sorting which of the infinite combination of models, linear, splines, polynomial or other models gives the best results. There are several challenges to overcome with prediction models. Some predictors are not correlated to the IMBD score in any way, some data contain outliers that throw off predictions and other real-world biases that may or may not be present in the dataset prevent model accuracy. Several models will be tested against the statistical methods and skills learned through the class to determine the best fit and most insightful model. With some hard work, a bit of luck and witchcraft, the final abracadabra will provide the most accurate prediction.

DATA DESCRIPTION – SETTING THE SCENE

The prediction variable or dependent variable is the IMDB rating. While the data set is large, some variables are not relevant to the prediction model and are therefore removed from the data set. *movieTitle, movieID,* and *imdbLink* are objective identifiers designed to keep track of the movies with a simple and unique easy identification. While useful for identifying a movie these variables have no measurable prediction power over the rating therefore, they were removed from the model.

The dependent variable, IMDB rating, demonstrates a left skew in the data (Appendix A) compared to a normal distribution. Therefore, most scores tend to be higher than a normal distribution mean value. The median score for a movie in our training data is 6.6 and the mean is 6.51 with a standard deviation of 1.1 .The boxplot shows that scores below 4 are considered outliers (Appendix B). After investigating the

dependent variable at the heart of the crystal ball the team focused on the independent variables surrounding it.

The first independent variable of note is the movie budget, which ranges from $560 thousand to $55 million. The mean value is $20.97 million. There is a right skew in the variable which indicates a bias. Part of the skew is expected by the availability bias, as the data dictionary indicates that not all movies disclose their budgets. However, another factor is at play here. The value of money changes over time and the relative power of budgets does too. For instance, a $1 million budget movie in 1960 is not the same as a 2003 movie with a similar budget. Therefore, the team actualized the movies' budgets to their present values by applying the Net Present Value formula assuming an annual interest rate of 2%. This approach led to a decrease in the relevance of the movie budget predictor.

The second variable is the release date. The data separates the day, month, and year of release. The release day shows normal distribution. No outliers were detected. next, the release month. There was no significant correlation between the month and score. The highest average score is in December, followed by November. While the regression may be relevant (p score of 6.8 e^{-3}) the score estimate of .02 increased stars per month on its own will not be an effective predictor. Furthermore, the 12 movies predicted are all equally released in November. Similarly, the year predictor may not be helpful on its own. However, the year data may be useful in another way. Investigating the most recent data may reveal relevant popular trends such as superhero movies, or niche films may be captured within recent movies helping to train the model. The year data demonstrates a left skew as fewer movies have been released in recent years, including 2020 which is an outlier year due to the COVID-19 pandemic halting movie releases.

The duration, in turn, falls into a range between 37 minutes and 330 minutes. While this is a broad duration, the median value is 106 and the mean is 109.7 minutes long with a standard deviation of 21 minutes. The duration may be an important predictor of the score but many movies have comparable durations. The boxplot shows many outliers for movies above average length and some outliers on the shorter side (Appendix C).

To best understand the independent variables within this model, the team has produced a basic variable table. This table groups together relevant measures such as the average, range, and quantiles for each predictor (Appendix D). The table helps get a sense of the data at a glance. For instance, all movies have a score between 1.9 and 9.3 so the 12 new movies are expected to fall within this range.

To push the data discovery further the team created box plots for every variable. These boxplots show issues with the outliers. Common outliers for each variable were identified with the Bonferroni outlier test and by creating studentized residual plots. Each of the independent variables in the matrix was plotted against IMDB Score separately and together (Appendix E). The outlier test identified the following movies as outliers: “Star Wars: Episode IV - A New Hope”, ‘Who’s Your Caddy?”, “Glitter”, “Disaster Movie”, and “Battlefield Earth”. The outliers will be removed from the dataset to further improve model accuracy and predictive power (Appendix F).

The relationship between quantitative predictors is a crucial factor in selecting and building a predictive model. Many different phenomena and effects like collinearity can throw off a model's predictive power. Understanding the relationship between variables is crucial. A preliminary correlation matrix reveals the degree of correlation between variables (Appendix G). Most of the variables do not show a high level of correlation with one another, but it is interesting to note that actor 2-star meter and actor 3-star meter show a .3 correlation. Further budget and duration show a .19 correlation. The team created a new predictor *avg_actor_rating* to calculate the average rating of the three main actors for each movie. This metric gave better results than the star meters separately. All the quantitative variables relationships are well below the .8 rule of thumb for removing variables due to collinearity. Therefore, no collinearity was noted within the dataset.

Next, the team investigated the categorical variables in the dataset. Some of the variables had more than a hundred distinct variables - such as cinematography, production company, director, and distributor company. We decided to first redefine categories manually, and then proceed with dummification or preprocessed data. While dummifying the variables is possible, the large increase in variables reduces the degrees of freedom therefore decreasing the predictive power. For the first step, excel pivot tables applied to each group to identify variations and amplitudes, frequency of local maxima, groups with similar distributions and trends. This information allowed the team to recategorize our parameters. For the second step, the best approach was to group the values with the highest numbers of movies and compare them with others. The dummification will be discussed more thoroughly in the modeling section.

Last, the team further explored the variables by checking for linearity with our decision variable (IMDBScore) and performed linear regressions for the relevant variables ie. *moviesbudget*, *duration*, *releaseYear*, *nbNewsArticles*, *country*, *maturityrating*, *nbFaces* and *movieMeter_imdbpro*. All variables indicated heteroskedasticity (with a p-value less than 0.05) except for the present value *movieBudget*.

MODEL SELECTION – THE MAGICIANS TRICK

First to construct a quality crystal ball, one needs a pure dataset. The team performed a generalized cleaning on the data. The first step is to eliminate collinearity and outliers. The Bonferroni test spotted outliers to remove from the dataset. Then the team detected and corrected heteroskedasticity within all the variables except for *movieBudget.* As part of the last data cleaning steps the team divided the categorical variables into top performers used as references, while the others were gathered in a variable 'Others'.

Next, the team explored the relationship between variables. Simple data visualization and statics gave insights to build on for the given dataset (Appendix H). A normal distribution of many variables led the team to believe that a series of linear regressions may be helpful to demonstrate the interdependence and effect of the variables on each other. However, the Tukey test for nonlinearity showed that only the *budget* could have a linear relationship with the *IMDBScore*; all other variables were non-linearly related to the score. Therefore, movie budget was kept as a linear predictor while other variables were explored.

To simplify model exploration the team investigated two separate models to determine the importance of independent variables. Reducing the number of independent variables can reduce complexity and positively impact the accuracy of the predicted *IMDBScore.* First, a random forest script assessed the important features for the analysis, the team chose this approach as it accounts for linear and non-linear models. The preliminary results give insight into the relevant predictors. Then, a package called *boruta* ranked the importance of the data features.

By combining both of these approaches it was clear which predictors are important and which to drop (Appendix I). The important factors are *actor1_starMeter, actor2_starMeter, actor3_starMeter, actor1, actor2, actor3, aspectRatio, maturityRating, language, country, director, cinematographer, distributor, releaseDay, plotKeywords, musical, western, sport,* and *productionCompany*. The most relevant features for the analysis are: r*eleaseYear, duration, language, moviemeter_IMDBpro, movie budget* (Appendix J).

Once the relevant fields picked, the team added some interactive terms to increase the predictive power of the model. These added terms represent the interaction between distinct variables. For instance, many films are action and adventure, not simply one . By combining these terms the model can accurately predict action-adventure movies which have a different average score. The final model consisted of the following variables: *movieBudget, releaseYear, nbNewsArticles, duration, matRating, director, drama, romance, horror, thriller, action-adventure* (interactions), *avg_actor_rating, animation, lang, scifi, nbFaces, war, crime.*

To find optimal polynomial fits, the team tested polynomial regressions for numerical parameters and calculated the minimum mean square error(MSE) as this model is 0.6061. the MSE is how far, on average, each observation is away from the line of best fit. The MSE of 0.6061 was for a polynomial of degree 9 for each of *movieBudget*, *releaseYear*, *nbNewsArticle* and *duration*. The team compared these results to spline regressions for different knots and found the combination of knots that resulted in the lowest MSE for the model. The final model was obtained by modeling the numerical variables *movieBudget, releaseYear, nbNewsArticles, duration, movieMeter_IMDBpro, release_month_num* as spline regressions with degrees 2, 2, 6, 3, 3 and 1 respectively along with the categorical variables and the interaction between them previously mentioned, with an MSE of 0.5929.

Lastly, to validate the results the team ran a cross validation test on our model (430) and obtained a minimum MSE of 0.468 (Appendix K). We eventually tested our model to predict the test movie ratings based on the features *Maturity Rating, Country, Language, Director, Cinematographer, Production company, and Distributors.*

RESULTS - ABRACADABRA

Foundations of predictions

After crafting the finest magic crystal ball, from mathematical models, variables and factors the model made predictions on new releases. The model ranked the to-be-released movies in the following order: *The Fablemans* 7.6, *Devotion* and *Strange World* at 6.5 each, *Falling for Christmas, Spirited* at 6.4 and *Blues Big City Adventure* at 6.4, *Slumberland* 6.2, *¡Que viva México!* and *The Menu* at 6.0, *Black Panther: Wakanda Forever* 5.2, *Poker Face* at 5.0 and finally, *Paradise City* at 4.9. The factors of each of these movies contributed to their final standing, however, to understand them, one must first understand the significance that the model places upon each predictor.

Multiple coefficients of very high significance levels are denoted by *** or **. All of the coefficients for the movie budget predictors showed strong levels of significance. Of note, all the movie budget coefficients were negative meaning that the higher the budget generally the lower the score of the movie, all else equal. As will be demonstrated, this idea of all else being equal will become difficult to discern. It seems counterintuitive that a high movie budget may be a negative predictor of the rating. However, a partial explanation is that a highly rated director or actor may command a sizable chunk of a film's budget and is forcing a high budgetary constraint. Therefore, the money may be spent on one aspect of the movie to the detriment of the rating.

The year that the film is released is also a significant predictor of ratings. The year shows a very significant p value. Therefore the predictor is likely directly negatively affecting the ratings. The coefficient is negative indicating that newer movies tend to have lower ratings. While year can be an important predictor nuance is crucial. The 12 predicted films are 2022 films, and the data set does not contain any 2022 data which may break from the established trend. Movies released during 2020 and 2021 had lower ratings potentially related to the height of the COVID pandemic.

The significance of the number of news articles is most significant at the third and fourth degree, where both receive the highest p score significance score. This is a positive coefficient so the more news articles that are released about a film the higher the score is anticipated to be. A limitation of note, is around the news articles. As the films are released at different times during the month of November, there may be more articles put out for the films that come out sooner rather than at the end of the month. The marketing push that corresponds with releasing a move may not yet be in full swing for the end of the month releases.

The duration of the film shows a strong significance for the negative coefficient in the second degree. The longer that the film is, the lower it is ranked. The IMDB movie meter also shows a significant negative coefficient at degrees 2,3,4. This indicates that for certain segments of movies that do not follow the general trends and categories as per the IMDB Pro meter, getting a higher meter score would mean a lower rating.

The next predictors, include the ratings of the films and most of the high output directors (director who have enough movies out that they can form a reasonable predictor, do not show to be significant predictors). Even though Allen as a director has a significant positive coefficient, none of the 12 upcoming films are directed by Allen. Spielberg does have a film coming out and his movies do have a positive coefficient, however the p value of this coefficient was .149 so well above a small enough p value to be significant.

When it comes to the genres, horror and action both reached 3-star significant levels. Both are negative coefficients however; the horror coefficient is twice as large as the adventure coefficient. The romance genre has a ** significance level and a negative coefficient. Drama and animation movies usually score better than other movies by roughly 0.5 IMDBScore. This is important as 2 of the 12 movies are animated: *Blues Big Adventure* and *Strange World*. On the contrary, horror and action movies might drop in rating compared to other genres if they had all other attributes similar.

To factor the actor ratings into the model the actor ratings were averaged. This allowed for all three of the main actor roles to be integrated into one predictor and one coefficient. This positive coefficient shows that when the average rating for the actors is higher, so will the score of the movie.

Results' interpretation

Looking at *The Fablemans*, the highest predicted film, you can see how it benefits from these coefficients in comparison to the other movies. *The Fablemans* falls in the bottom half of the movies when ranked by budget so it is less negatively impacted by the budget coefficient than the comparable films are. *The Fablemans* is the third longest film, which also holds a negative predictor. It does rank second in the number of news articles and fifth best for the average actor score which lends another two positive coefficients to its projected ranking. *The Fablemans* also did have the most faces in its poster which is another negative coefficient, however it is evidently one that it is able to overcome. While not one of our significant predictors, the movie also does benefit from having Speilberg as the director as he is one of the prolific directors that maintains a positive coefficient in this model.

On the other end of the continuum is *Paradise City*. While it has the third lowest budget and is the second shortest movie it has the third fewest news articles written about it. Further it falls into the bottom half of the averaged star power and it falls into the top half of releasing films for budget and third in the number of faces in the poster. The combination of these factors make it so our model has little confidence in the success of this movie.

Looking at the results for all the films, but especially the best in *The Fablemans* and the worst in *Paradise City* it is clear that there are many different areas that the films have to balance. It is not sufficient to simply shine in one category when this predictive model factors in so many different aspects of the to be released movies. It is also a testament that the highest score was still a mediocre score of 7.6. This speaks to the difficulty in producing the time of a movie that is seen as a masterpiece, even for a director who has done it before like Speilberg. If these projections will prevail as true, only time will tell, however the impossibility of checking all the right boxes in this industry is something that cannot be ignored.

Even though the model is looking at specific predictions for 12 movies, various statistical parameters measure the predicting power of the model as a whole. One important measure is the R^2 coefficient. R^2 describes the amount of variation that the independent variables explain for the dependent variable. An R^2 score of 1 indicates that the independent variables explain perfectly all the variance of the dependent

variable, whereas an R^2 score of 0 indicates that the independent variables do not at all explain the dependent variable.

This model's R^2 score is 0.52. In practical terms, this indicates that the variables in the dataset explain 52% of the variance in the *IMDB rating.* While 52% may seem quite low, relative to other models this is reasonable. Some literature reports on much lower values or claims significance in r scores below this number. [1,2] Of course there is no standard r score as each model is defined within its own context. This model, for instance, looks at predicting the feelings or opinions towards something as complex as art and media on a mass scale. In some cases, a breakout movie or social phenomena comes from nowhere. The model places a high emphasis on a budget but a movie like *Reservoir Dogs* with a small budget and cast would be an outlier that exceeds expectations. It is rare that social phenomena as complex as emotions, social phenomena, culture and feelings are deterministic without knowing all the factors or measuring the interconnected nature of each contributor. This model therefore leaves a simple question: What factors make for a great movie?

Limits and conclusion

The model may only be able to answer 52% of that question but a second statistic demonstrates the model's true potential. The error associated with regressions is commonly measured as a mean square error (mse). This statistic measures how far, on average, each observation is away from the line of best fit. In other words, on average how many stars is the model away from the actual movie's rating. The mse for the model is on average around 0.55 stars. put simply, on average the model can predict within 0.55 stars of the "true" value in this data set. This is a relatively tight range that allows for robust predictions of the data set. When predicting the score of the 12 new movies we can safely say that on average we will predict their scores within a range of ±0.55 stars. While the model requires a more open interpretation there is enough confidence to predict a bad, okay, good, great or excellent movie.

It is possible to increase the statistical values of the mse and the R^2 coefficient, however, this would ruin the model. One of the dangers in data science is overfitting the data with a model. This means that the model is fitting every possible point and creating an arbitrarily complex function. There is no inherent problem with overfitting the data, except when predicting new data. Since the model is arbitrarily complex it often cannot predict new data with any reasonable certainty. The model may have an mse of 0 and an R score of 1 but will predict a rating of -1 for *Blue's Big City Adventure* which is not a possible rating.

A common trick to avoid the risk of overfitting involves splitting out part of the data to test it against a model. In mathematics, a sample data set is generally representative of the larger population so long as the data is large enough and the sample is randomly selected. In reality, this predictive algorithm splits the data randomly 100 times through a pseudo-cross-validation looped function. The function first splits the data in a 70-30 percent random split then test the model, then this process is repeated 100 times. The pseudo cross-validation loop investigates and creates a model based on the 70% of data it has access to tests the accuracy against the remaining 30% of the data. For each of these 100 iterations, the statistical data is measured and compiled. The best model balances both the mse and R score statistics as both are critical ingredients to a successful model.

In the end, something as complex as predicting emotions and review ratings cannot be boiled down to a few variables with some predictions. The crystal ball of the future relies on simple tricks. While the model is a valiant attempt, more data and an understanding of complex relationships within the society and framework in which the movies are shown are required for more predictive power and accuracy. This model stands as a great predictor of the future.

APPENDIX & WORKS CITED

Works Cited

(1) Chew, N. W.; Lee, G. K.; Tan, B. Y.; Jing, M.; Goh, Y.; Ngiam, N. J.; Yeo, L. L.; Ahmad, A.; Khan, F. A.; Shanmugam, G. N. A multinational, multicentre study on the psychological outcomes and associated physical symptoms amongst healthcare workers during COVID-19 outbreak. Brain, behavior, and immunity 2020, 88, 559-565

(2) Lowenkamp, C. T.; Bechtel, K. The predictive validity of the LSI-R on a sample of offenders drawn from the records of the Iowa Department of Correction Data Management System. Fed. Probation 2007, 71, 25.

Appendix A

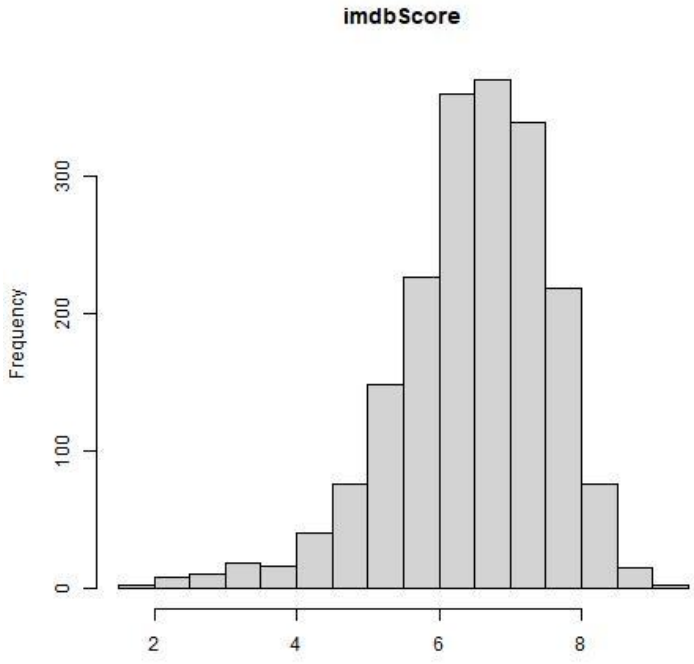


Figure 1 Histogram of ratings, demonstrating a left skewness of ratings.

Appendix B

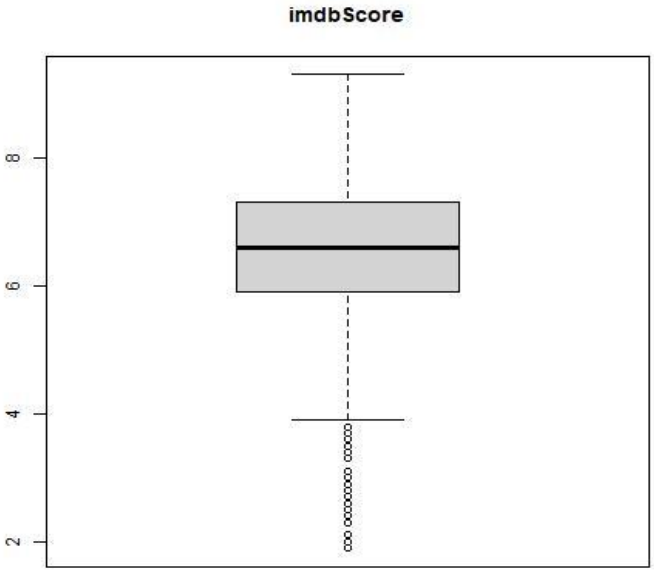


Figure 2 Boxplot of IMDB scores

Appendix C

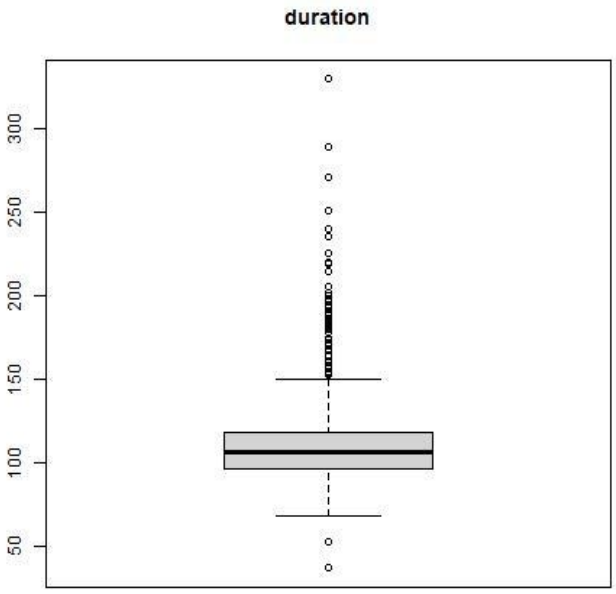


Figure 3 Boxplot of Duration

Appendix D

Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Pctl. 25	Pctl. 75	Max
imdbScore	1930	6.512	1.1	1.9	5.9	7.3	9.3
releaseDay	1930	15.952	8.304	1	9	23	30
releaseYear	1930	2001.392	11.776	1936	1997	2010	2018
duration	1930	109.699	21.246	37	96	118	330
aspectRatio	1930	2.096	0.273	1.18	1.85	2.35	2.76
nbNewsArticles	1930	770.609	1860.425	0	78	845.5	60620
actor1_starMeter	1930	21189.557	285993.418	9	505	4665	8342201
actor2_starMeter	1930	17114.095	173281.578	3	1895.25	7667	5529461
actor3_starMeter	1930	35468.581	260425.782	8	3075	12250.5	6292982
colourFilm	1930						
... Black and White	63	3.3%					
... Color	1867	96.7%					
nbFaces	1930	1.44	2.06	0	0	2	31
action	1930	0.201	0.4	0	0	0	1
adventure	1930	0.126	0.332	0	0	0	1
scifi	1930	0.108	0.311	0	0	0	1
thriller	1930	0.298	0.457	0	0	1	1
musical	1930	0.07	0.256	0	0	0	1
romance	1930	0.245	0.43	0	0	0	1
western	1930	0.018	0.132	0	0	0	1
sport	1930	0.048	0.214	0	0	0	1
horror	1930	0.113	0.317	0	0	0	1
drama	1930	0.549	0.498	0	0	1	1
war	1930	0.036	0.187	0	0	0	1
animation	1930	0.01	0.101	0	0	0	1
crime	1930	0.216	0.412	0	0	0	1
movieMeter_IMDBpro	1930	11612.381	40181.21	71	2836.25	10198	849550
movieBudget	1930	31.287	22.1	0.824	13.154	45.61	126.289

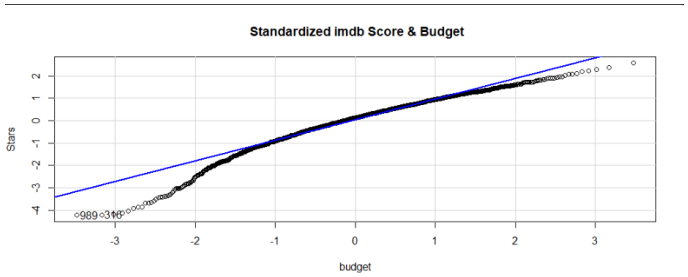
Figure 4 Summary Statistics

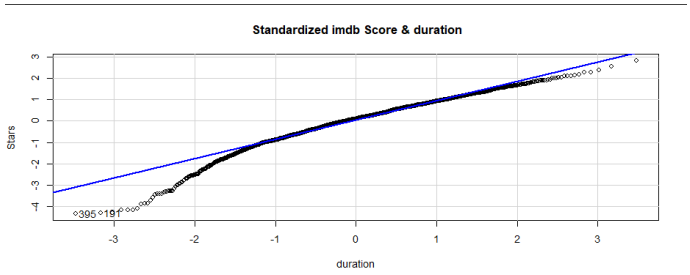
Appendix E

	movieBudget	duration	nbNewsArticles	actor1_starMeter	actor2_starMeter	actor3_starMeter	nbFaces
movieBudget	1	0.19	0.03	-0.02	-0.03	-0.03	0.03
duration	0.19	1	0.09	-0.004	0.03	-0.01	0.01
nbNewsArticles	0.03	0.09	1	-0.02	-0.02	-0.03	-0.03
actor1_starMeter	-0.02	-0.004	-0.02	1	0.18	0.04	-0.002
actor2_starMeter	-0.03	0.03	-0.02	0.18	1	0.30	-0.01
actor3_starMeter	-0.03	-0.01	-0.03	0.04	0.30	1	-0.005
nbFaces	0.03	0.01	-0.03	-0.002	-0.01	-0.005	1

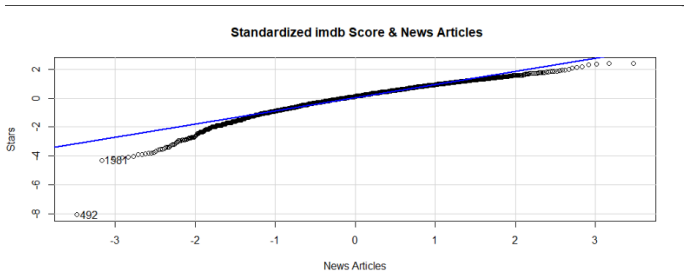
Figure 5 Correlation Matrix

Appendix F

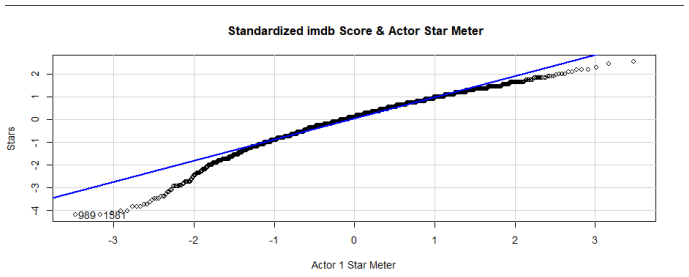




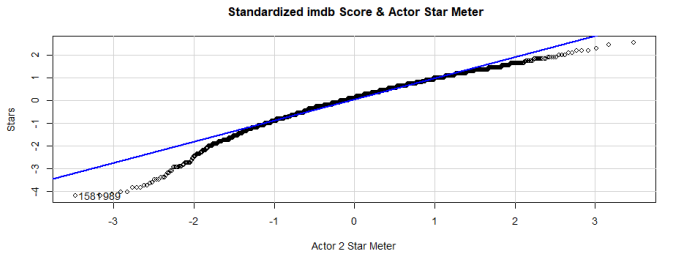
Duration Outliers: 395, 191, 1806



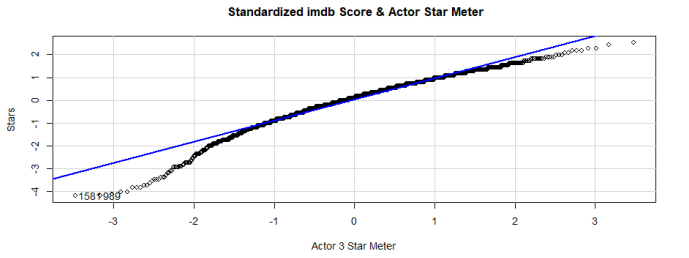
News Articles Outliers: 492, 1581, 989



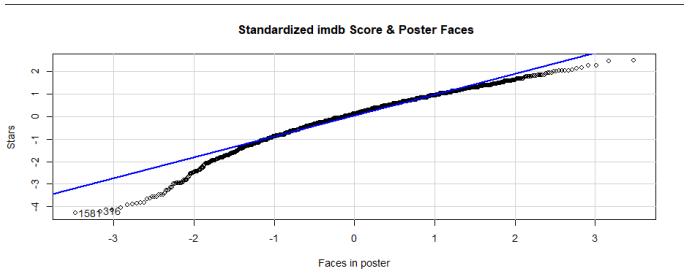
Actor 1 Outier: 989



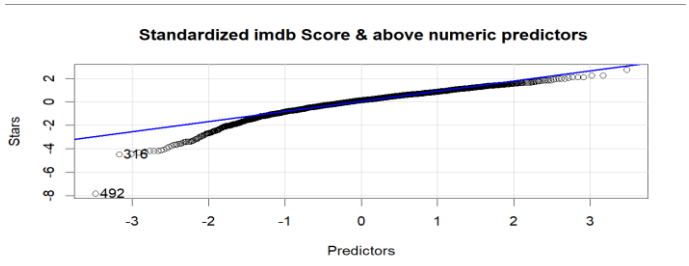
Actor 2 Outlier: 1581



Actor 3 Outlier: 1581



Outlier: 1581



Outliers: 492, 316, 1806, 1581, 191

Figure 6 Base Standardized plots, looking for outliers

Appendix G

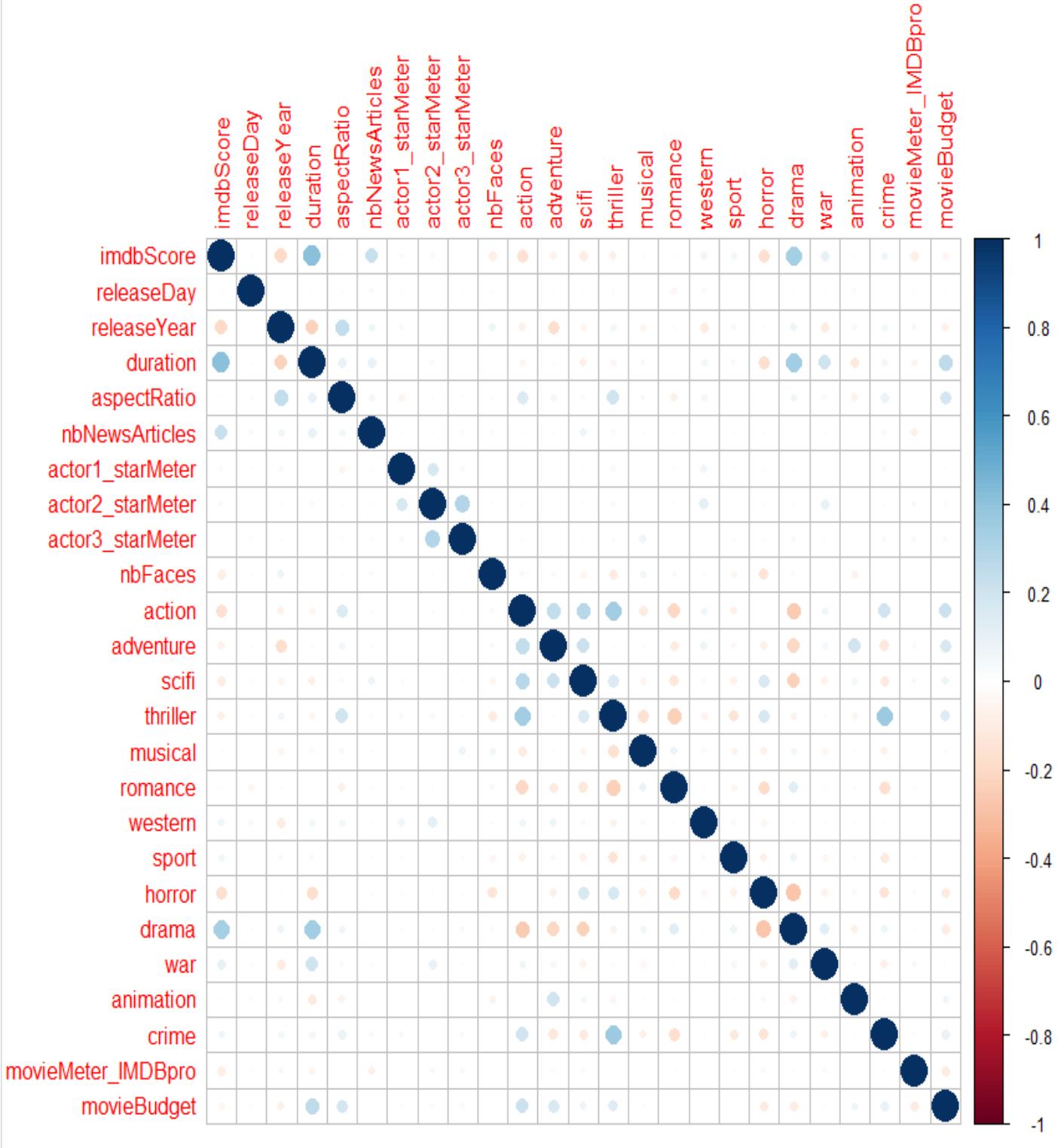


Figure 7 Correlation Matrix

Appendix H

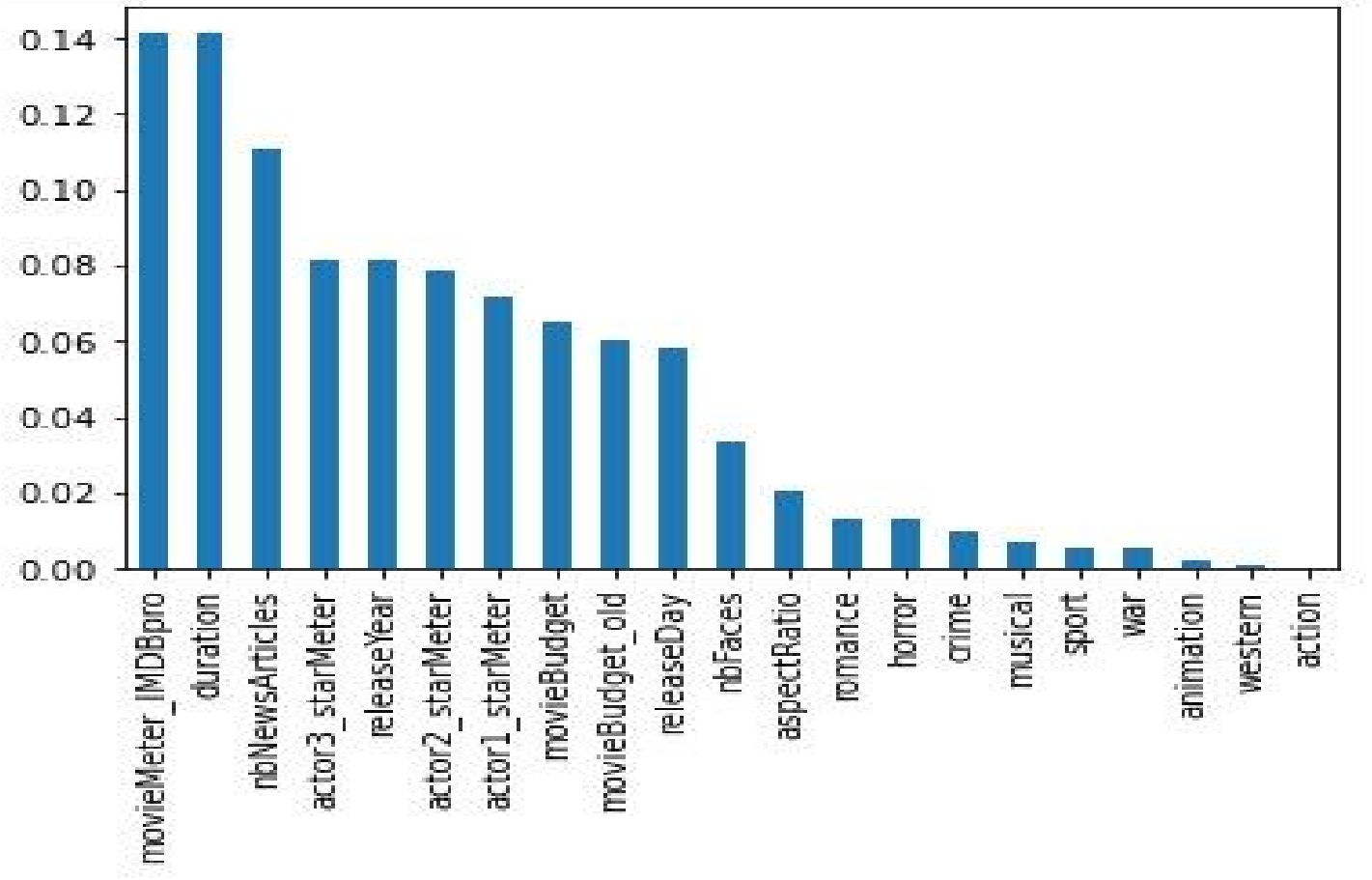


Figure 8 Histogram of predictors

Appendix I

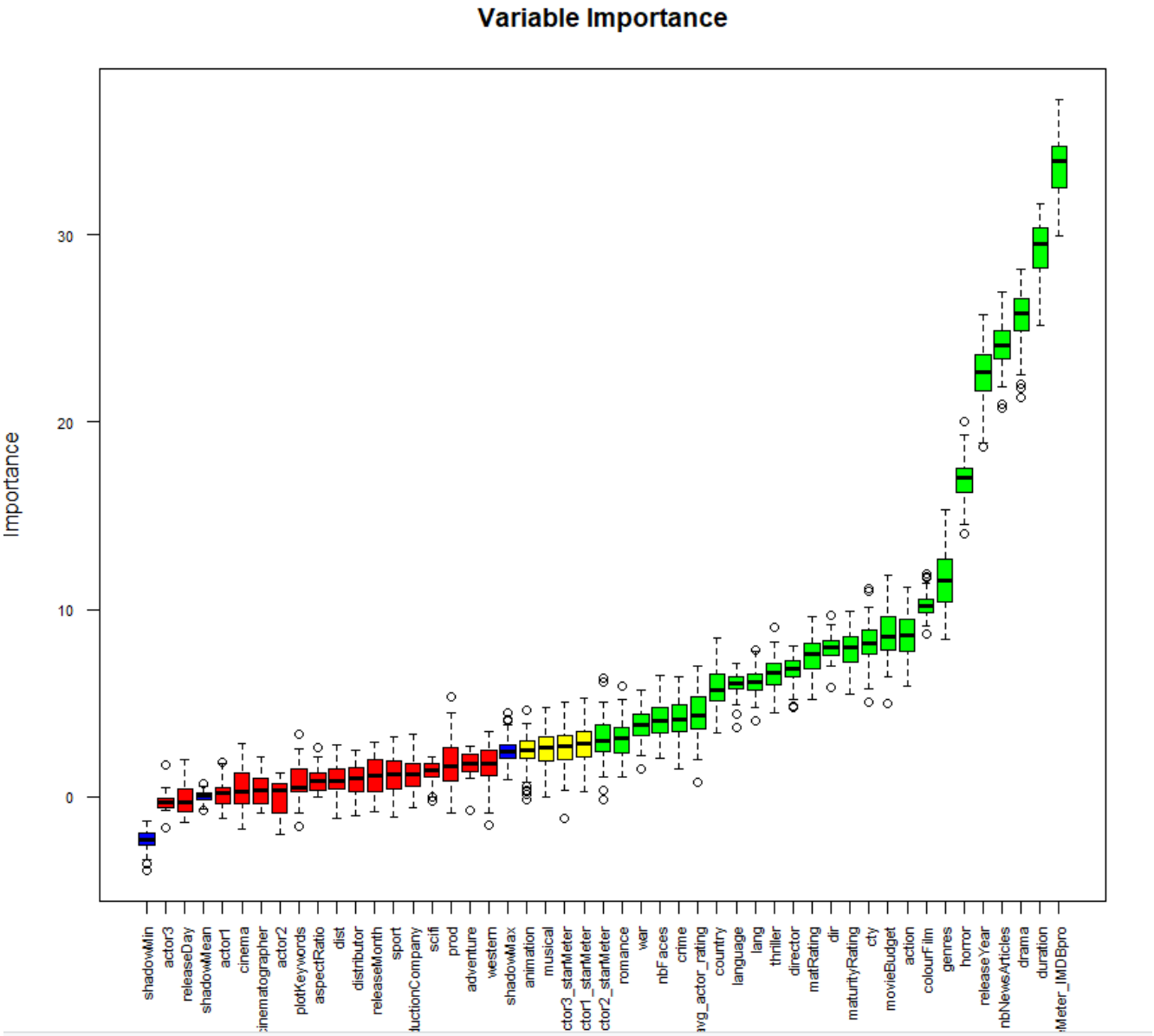


Figure 9 Boruta Feature Selection Outcome

Appendix J

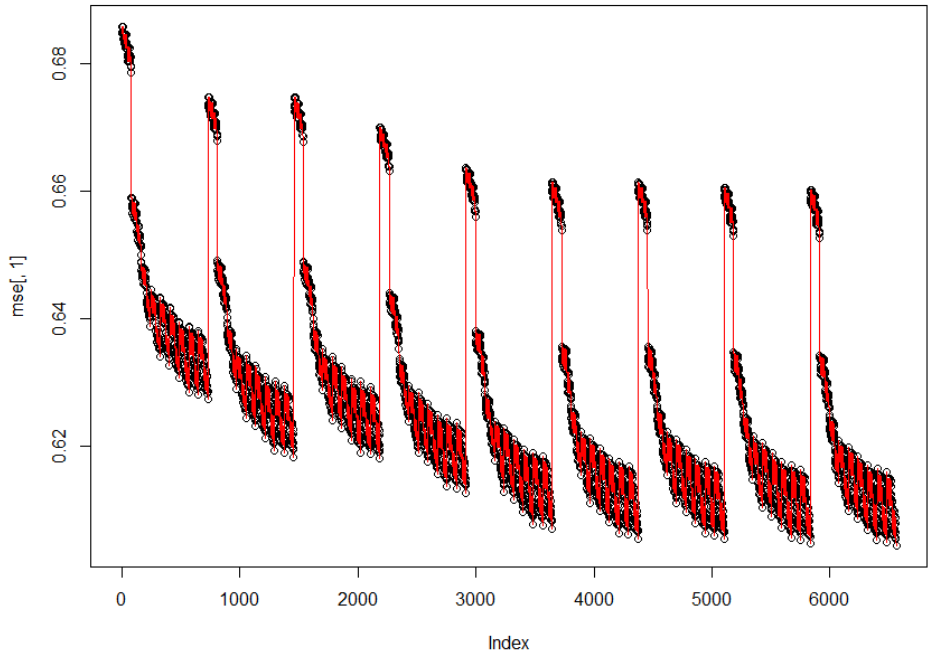


Figure 10 Relevancy of Predictors

Appendix K

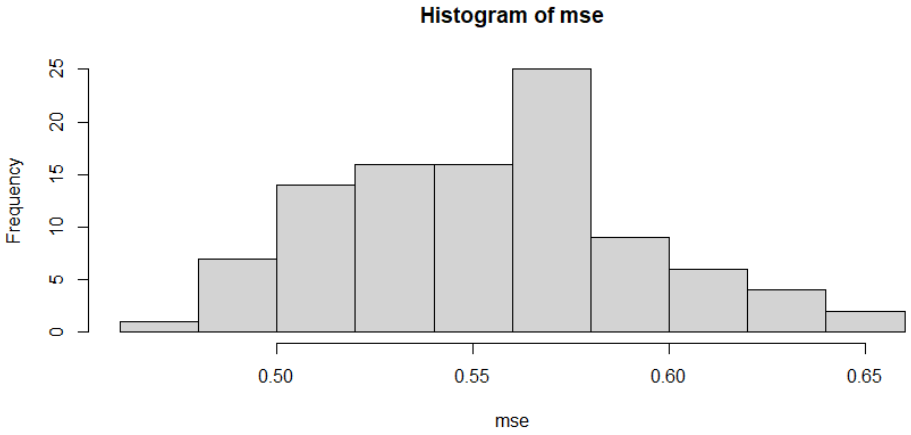


Figure 11 MSE distribution for final model cross validation